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Altered oscillatory neural dynamics related to word prediction in older adult readers

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ABSTRACT

The engagement of predictive mechanisms during language comprehension can facilitate processing and modulate neural oscillatory activity. These modulations include alpha-band activity decreases prior to expected words, reflecting anticipatory preparation, and frontal theta-band activity following unexpected words, reflecting engagement of cognitive control. It remains unknown how these oscillatory dynamics are impacted by aging. Here, we re-analysed previously collected EEG data from a sentence reading experiment with older adult participants to examine oscillatory responses to expected and unexpected words. Compared to younger adults, older adults' frontal theta responses following unexpected words were delayed and reduced in magnitude. Alpha-band decreases prior to expected words were comparable across age groups, but only young adults' pre-stimulus alpha activity was correlated with the post-stimulus frontal theta response. These results suggest older adults may engage anticipatory neural mechanisms comparably to younger adults, but do not engage the same control mechanisms following prediction errors.

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Language; prediction; aging; EEG; oscillations

Introduction

Despite the seemingly relative ease with which we understand written and spoken language, the successful comprehension of linguistic stimuli involves the engagement of numerous neurocognitive processes to construct meaning from a stream of noisy input. For instance, the comprehender must not only engage sensory processing mechanisms to derive meaning from stimuli, but also build and continually update a message-level representation, or situation model, with incoming new information (Zwaan et al., 1995; Zwaan & Radvansky, 1998) in order to construct a message from a string of individual words. However, the comprehender can also use the accumulated contextual information provided by the linguistic stimuli, as well as the mentally constructed situation model, to generate predictions about potential upcoming words in order to facilitate processing in the moment (Kuperberg et al., 2020; Pickering & Garrod, 2013). Indeed, a large body of empirical evidence, including behavioural, eye-tracking, and neural results, supports the notion that comprehenders actively generate features of upcoming words in order to more rapidly process incoming language

(Altmann & Mirković, 2009; Federmeier, 2007, 2022; Kuperberg & Jaeger, 2016; Pickering & Gambi, 2018). This research suggests that many different features of upcoming words, including orthographic, phonological, and morphosyntactic features, can be predicted (Altmann & Kamide, 1999; Arai & Keller, 2013; DeLong et al., 2005; Dikker et al., 2010; Kim & Lai, 2012), and that this prediction process may be engaged rapidly following a preceding word or stimulus (Hubbard & Federmeier, 2021; Wang, Kuperberg et al., 2018). Prediction may play an important role in the comprehension of rapidly incoming language stimuli, and thus understanding the neural mechanisms underlying predictive processing is of critical importance.

To study the mechanisms underlying prediction, many cognitive neuroscientific investigations of language processing have used non-invasive neuroimaging methods to measure changes in brain responses as participants comprehend linguistic stimuli. These include studies examining how components of the event-related potential (ERP), or the electrophysiological response time-locked to the onset of a stimulus, differ in amplitude based on the predictability of words that are

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read. For instance, numerous studies have demonstrated that the N400, a centroparietal negativity reflecting retrieval of information from semantic memory (Kutas & Federmeier, 2000, 2011), is sensitive to the predictability of its eliciting word (Federmeier et al., 2007; Federmeier & Kutas, 1999; Szewczyk & Schriefers, 2018; Wlotko & Federmeier, 2012a), suggesting prediction (among other processes) can facilitate semantic access. Additionally, contextually unexpected or semantically anomalous words in sentences can modulate the amplitude of post-N400 ERP components, including anterior and posterior positivities, potentially reflecting revision of the situation model to incorporate the unexpected information (Brothers et al., 2020; DeLong et al., 2014; Kuperberg et al., 2020; Van Petten & Luka, 2012). These results indicate that the brain is sensitive to the predictability of information when processing language, and engages specific mechanisms after encountering unpredictable words.

Although the approach of investigating differences in ERP amplitudes has led to several key insights into the mechanisms underlying prediction, examination solely of ERP components leaves out other important neural signals – in particular, rhythmic or oscillatory brain activity. Neuronal oscillations, or rhythmic synchronisation and desynchronisation of activity at particular frequencies, are of critical importance for communication of information across areas of the brain, and may support many cognitive processes (Bauer et al., 2020; Buzsaki & Draguhn, 2004; Ward, 2003), including language comprehension and production (Meyer, 2018; Peelle & Davis, 2012; Piai & Zheng, 2019). Neural oscillations may additionally play a role in supporting anticipatory mechanisms when comprehending language (Arnal & Giraud, 2012; Lewis & Bastiaansen, 2015). By applying time–frequency analysis methods to EEG data (Cohen, 2014) collected from individuals reading linguistic stimuli that vary in their predictability, changes in neural oscillatory power at different frequencies can be related to predictive processing during language comprehension.

Prior work utilising these methods has identified changes in neural oscillatory activity prior to the onset of a predictable word, potentially suggesting engagement of anticipatory processing. More specifically, greater decreases in alpha / beta band activity (roughly 8–25 Hz), generally over occipital sensors, have been observed prior to a highly predictable word compared to an unpredictable word (León-Cabrera et al., 2022; Li et al., 2017; Rommers et al., 2017; Terporten et al., 2019; Wang, Hagoort et al., 2018). Interestingly, decreases in alpha band activity are also observed prior to the *production* of more predictable words (Gastaldon

et al., 2020; Piai et al., 2014, 2015), in line with the notion that prediction of upcoming words involves similar mechanisms involved in planning and production of words (Dell & Chang, 2014; Pickering & Garrod, 2013). Additionally, similar power decreases in this frequency range have been observed preceding more predictable stimuli during spatial (Capotosto et al., 2009; Gould et al., 2011) and temporal (Rohenkohl & Nobre, 2011) prediction tasks, and are also observed when attention is oriented to anticipate upcoming visual stimuli (Foxe et al., 1998; Worden et al., 2000) suggesting that a more domain-general process may underlie prediction of upcoming stimuli. These results may be in line with a predictive coding account of neuronal activity (Friston & Kiebel, 2009), which posits that oscillations in the alpha / beta frequency range reflect anticipatory top-down control and regulation of the bottom-up input (Bastos et al., 2020; Friston et al., 2015). Indeed, it has been recently suggested that beta oscillatory activity observed during sentence comprehension reflects the propagation of predictions of sentence representations to lower levels of the processing hierarchy (Lewis et al., 2016; Lewis & Bastiaansen, 2015).

Changes in neural oscillatory power have also been observed in response to unpredictable words, or prediction errors. These responses have primarily been in the form of increases in theta band power (roughly 3–8 Hz), often over frontal sensors, following unpredictable words or semantic violations (Bastiaansen & Hagoort, 2015; Hagoort et al., 2004; Hald et al., 2006; Kiehl et al., 2014, 2015; Rommers et al., 2017; Wang et al., 2012), as well as following grammatical violations or unexpected sentence structures (Lewis et al., 2023). Increases in frontal theta band activity have also been observed in response to unexpected events or stimuli (Hajihosseini & Holroyd, 2013; Mas-Herrero & Marco-Pallarés, 2014), as well as response errors (Cohen, 2011; Luu et al., 2004; Luu & Tucker, 2001), in other tasks not involving language processing. These results are in line with the notion that the frontal theta signal may reflect a more domain-general cognitive control process (Cavanagh & Frank, 2014) that is recruited following errors or unexpected events. Importantly, there is evidence that the previously discussed pre-stimulus alpha-band decreases and the frontal theta-band increases may be linked, with the magnitude of alpha desynchronisation predicting frontal theta synchronisation, suggesting that anticipatory processing can influence the engagement of cognitive control following a prediction error (Kaiser et al., 2022; Rommers et al., 2017). In sum, during language comprehension, decreases in alpha-band activity prior to a predictive stimulus may reflect anticipatory processing, and frontal theta-band activity

following unexpected words likely reflects the engagement of cognitive control mechanisms following a prediction error.

Understanding the neurophysiological and oscillatory mechanisms underlying prediction during language comprehension in healthy younger adults can provide insights and hypotheses for developing studies to examine how these mechanisms might differ in populations that show impairments in engaging prediction. For instance, research suggests that older adults, despite showing preserved processing in many areas of language comprehension (Park et al., 2002; Park & Reuter-Lorenz, 2009), do not always receive the same facilitation for predictable words when comprehending language, particularly when language input is rapid (Janse et al., 2007; Kliegl et al., 2004). In particular, ERP studies have demonstrated that N400 amplitudes are less sensitive to predictability in older adults compared to young adults, and that older adults do not elicit post-N400 responses following prediction errors to the same degree as young adults (Dave et al., 2021; Kutas & Iragui, 1998; Wlotko et al., 2010, 2012; Wlotko & Federmeier, 2012b). These results suggest older adults may not engage predictive processing mechanisms to the same degree as younger adults when comprehending language. One possibility is that older adults have greater difficulty constructing a message-level representation from the preceding sentence context, or using this context to guide their predictions (Federmeier & Kutas, 2005; Payne & Federmeier, 2018). However, another possibility is that some aspects of predictive processing are relatively preserved in older adult comprehenders, but older adults do not show the same engagement of control or revision processes following unpredictable words (Dave et al., 2018). An examination of neural oscillatory dynamics related to word prediction in older adult readers could provide insight into which specific mechanisms are impaired in older adults compared to young adults; however, no study to date has made this comparison.

Impairments or alterations in the engagement of neural oscillatory mechanisms in older adults have been reported in other related research. For instance, some studies have shown reduced engagement of frontal theta-band activity following errors or unexpected stimuli in older adult participants compared to young adults (Kolev et al., 2005, 2009; Tafuro et al., 2019; Van de Vijver et al., 2014), suggesting older adults may not engage cognitive control mechanisms to the same degree as younger adults. In contrast, studies examining how alpha-band activity decreases prior to predictable stimuli differ between age groups have reported mixed results, with some studies

suggesting a preservation of anticipatory processing across age (Heideman et al., 2018; Zanto et al., 2011). An important caveat is that these studies primarily employed more simplistic tasks, and were not specifically focused on oscillatory dynamics during language comprehension, which may to some degree be preserved with age.

In the present study, we aim to fill this gap in the literature by analysing the spectral neural dynamics during language comprehension in older adult readers and comparing these results to those from young adult readers comprehending the same stimuli. To do so, we re-analysed two previous datasets, in which individuals read sentences while their EEG was recorded, using time–frequency analysis methods: data from Federmeier et al. (2007), which had young adult participants, and Wlotko et al. (2012), which had older adult participants. Importantly, the stimuli and procedure of these two datasets were identical, allowing us to make tightly controlled comparisons across age groups. In these studies, the sentences that individuals read were either strongly or weakly constraining towards a particular ending word, and ended with either an expected or an unexpected word. This design allowed for examination of not only pre-stimulus preparatory responses to more predictable stimuli (by comparing neural activity in strongly vs. weakly constraining sentences), but also post-stimulus responses specific to prediction errors (by comparing neural activity following unexpected vs. expected words).

Methods

This paper uses new methods to reanalyse the data from Federmeier et al. (2007), as well as Wlotko et al. (2012). The materials and procedures used in these studies were identical. Further specifics of the methodology can be found in these publications.

We note that the Federmeier et al. (2007) study was previously re-analysed with time–frequency analyses, the results of which were published in Rommers et al. (2017). Here, we once again re-analysed this data, but using different time–frequency analysis methods than in Rommers et al. (2017). More specifically, we used wavelet decomposition and a different baseline correction method to correct for potential bias caused by traditional correction methods (Ciuparu & Mureşan, 2016; Grandchamp & Delorme, 2011). Thus, we report the results of this analysis to (a) demonstrate that the results from Rommers et al. (2017) generalise across analytic choices and (b) directly compare with the older adult results. However, the young adult results are not novel to this paper.

Participants

The young adult participants from Federmeier et al. (2007) were 32 right-handed individuals at the University of California, San Diego who participated in the experiment in exchange for course credit or cash. One individual was dropped due to technical issues with importing the data, resulting in a total of 31 participants in the analysis. All participants reported normal or corrected vision and had no history of any neurological or psychiatric disorder. The mean age was 20 years (range 18–28 years), and 16 of the participants were female. The study was approved by the local ethics committee, and all participants provided written informed consent and were debriefed following participation.

The older adult participants from Wlotko et al. (2012) were 24 right-handed individuals from the local San Diego community who participated in the experiment in exchange for cash. The mean age was 72 years (range 59–88 years), and 12 of the participants were female. All participants reported normal or corrected vision and had no history of any neurological or psychiatric disorder. Additionally, all participants were screened using the Mattis Dementia Rating Scale (Mattis, 1976), and did not show cognitive impairment (range of 136–144 out of a maximum of 144; mean = 142).

We note that although the original publication describing the older adult data was published a few years later than the publication describing the younger adult data, these datasets were in fact collected in an overlapping timeframe (the younger adults in 2003–2004 and the older adults in 2004), by the same personnel, in the same lab, and using the same equipment.

Materials

The stimuli were comprised of 282 English sentence frames, half of which were strongly constraining (cloze probability of the final word > 67%; mean = 85.3%), and the other half of which were weakly constraining (cloze probability of the final word < 42%; mean = 26.9%). Half of the sentence frames were completed with the expected ending word (the word with the highest cloze probability for that sentence frame), while the other half were completed with an unexpected word (cloze probability < 2%) that was still plausible in the sentence context. Thus, the critical sentence ending words were divided into four categories: Strong Constraint Expected (SCE), Strong Constraint Unexpected (SCU), Weak Constraint Expected (WCE), and Weak Constraint Unexpected (WCU). The same unexpected ending words were paired with both a strongly constraining and weakly constraining sentence

Table 1. Examples of stimuli presented during the experiment.

Constraint	Sentence Frame	Ending	Ending Type
SC	Sam could not believe her story	true	Expected (E)
	was	published	Unexpected (U)
	The children went outside to	play	Expected (E)
WC	I was impressed by how much he	look	Unexpected (U)
		knew	Expected (E)
	Joy was too frightened to	published	Unexpected (U)
		move	Expected (E)
		look	Unexpected (U)

Note: SC = strongly constraining, WC = weakly constraining.

frame so that lexical properties of the unexpected conditions were matched across levels of constraint. Sentence presentation was counterbalanced, such that participants saw each sentence frame only once, but across participants each sentence frame was completed equally often with its expected and unexpected sentence ending.

Words immediately prior to the final word of the sentence (i.e. presented during the time that the pre-stimulus alpha power measurements were collected) were primarily made up of function words, determiners or prepositions ("the", "a", "his", etc.; 65% of pre-final stimuli). Examples of the sentence stimuli are presented in Table 1.

Procedure

Participants were seated approximately 100 cm from a CRT computer monitor in a sound attenuated and electrically shielded room. Following EEG setup, participants were provided instructions that they would be presented with sentences to read for comprehension, and that they would be asked questions about what they had read following the EEG session. They were asked to minimise blinks, eye movements, and muscle movement while reading the sentences. The participants were then presented with a set of practice sentences to acclimate the participants to the task situation and presentation format. Following the practice session, the experimental session began.

Each trial of the experimental session began with a warning signal (several pluses presented on the screen for 500 ms), indicating the trial was beginning. A blank screen followed the warning signal for a random period between 500 and 1200 ms. The presentation of the sentence then began. Sentences were presented word-by-word in a rapid serial visual presentation (RSVP) format in the centre of the screen. Each word of the sentence was presented for 200 ms, followed by a blank period of 300 ms. A 3 s pause separated sentences. The experimental session was divided into four blocks of

roughly 70 sentences each, allowing participants to take a short rest between each block. Overall recording time was approximately one hour.

EEG recording and pre-processing

EEG data were recorded from 26 Ag/AgCl electrodes embedded into a flexible elastic cap and distributed over the scalp in an equidistant arrangement. Additional facial electrodes were attached for monitoring of electro-oculogram (EOG) artefacts, including one adjacent to the outer canthus of each eye and one below the lower eyelid of the left eye. Electrode impedances were kept below 5 k Ω . Signals were amplified by a Grass amplifier with an online bandpass filter of 0.01–100 Hz, as well as a 60 Hz notch filter, and a sampling rate of 250 Hz. The left mastoid electrode was used as a reference for on-line recording; offline, the average of the left and right mastoid electrodes was used as a reference.

Following data collection, each raw EEG time series was filtered with a 50 Hz lowpass Butterworth filter with a 12 dB/oct roll-off. Low-frequency noise such as slow drifts was removed with a robust detrending method, in which a higher-order polynomial was fit to the data and subtracted out (de Cheveigné & Arzounian, 2018). The signal then was segmented into epochs from 2000 ms prior to stimulus onset to 1800 ms following stimulus onset for each sentence ending word. The large size of the epochs was for examining pre-stimulus activity, and having enough time to examine low-frequency oscillatory activity (Cohen, 2011). No baseline correction was performed at this stage. To correct ocular artefacts (eyeblinks and saccades), the data were decomposed into independent components with the Adaptive Mixture Independent Component Analysis (AMICA) algorithm (Hsu et al., 2018; Palmer et al., 2012), and the correlations between each independent component timecourse and the bipolar VEOG channel (lower eye channel – left prefrontal channel), as well as the bipolar HEOG channel (left outer canthus channel – right outer canthus channel) were calculated in order to find the component(s) containing artefacts. Components with a high correlation were removed, and the remaining components were then recombined to reconstruct the EEG data. The data were then scanned with an amplitude threshold (100 μ V threshold), and finally manually checked by the experimenter for any additional artefacts. In total, in the young adult group, an average of 3% of trials were removed, with a range of 0% to 14% across participants, and in the older adult group, an average of 3% of trials were removed, with a range of 1–8% across participants.

Time–frequency analysis

Time–frequency decomposition of the EEG data into oscillatory power over time was performed using Field-Trip functions in MATLAB (Oostenveld et al., 2011). The data in the EEG epochs were convolved with Morlet wavelets that varied in width (number of cycles) to balance temporal and frequency precisions across the range of frequencies. The series of Morlet wavelets used for the time–frequency transform ranged from 2 to 30 Hz in steps of 0.5 Hz, with a width that started at 3 and increased linearly to a width of 7 across the range of frequencies. Oscillatory power at these frequencies was extracted across time in 20 ms increments. Specific details regarding the temporal and spectral precision are reported in the Supplementary Materials. The time–frequency data was then baseline corrected with a concatenated z-score procedure that reduces potential biases from standard baseline correction procedures (Ciuparu & Mureşan, 2016), in which oscillatory data in a baseline period time was extracted from each trial, at each frequency and channel, and concatenated across trials. The mean and standard deviation of this concatenated baseline data were then used to z-score the trial data. For analysis of pre-stimulus activity, a baseline period of –1400 to –1200 msec (i.e. 400–200 msec prior to the onset of the second to last word) was extracted from each trial. We expected that pre-stimulus activity would differ based on predictability; however, this difference could influence the evaluation of post-stimulus activity if the pre-stimulus window was used as a baseline. Thus, we used the same baseline period of –1400 to –1200 ms when analysing post-stimulus activity. The baseline corrected trials were then averaged together.

The resultant time–frequency representations could be influenced by the underlying ERP (i.e. phase-locked activity) rather than just by distinct oscillatory mechanisms (i.e. non-phase-locked activity). To investigate this, we conducted an additional analysis in which the ERP was first subtracted from the data prior to the time–frequency transform, and the analyses were conducted on this data. These results did not differ from those we report below (without the ERP subtraction), and are thus described in the Supplementary Materials.

Statistical analyses of oscillatory data

To avoid potential issues of channel and time window selection leading to spurious significant results (Luck & Gaspelin, 2017), statistically significant differences in oscillatory power between conditions were identified with mass univariate statistics, corrected for multiple

comparisons using a non-parametric cluster-based permutation approach (Maris & Oostenveld, 2007). In these tests, the difference in average oscillatory power between conditions was calculated for each participant, and a dependent *t*-test was conducted at each time–frequency–channel bin, testing for a significant difference from 0 (a difference between conditions) across subjects. Significant *t*-values that were adjacent in space, time, and frequency were then clustered together. These clusters were characterised by taking the sum of *t*-values of the adjacent points within the cluster. To correct for multiple comparisons, the observed clusters were compared to a permutation distribution. This distribution was generated by shuffling the condition labels of the data, performing the *t*-tests on the shuffled data, finding clusters of significant *t* values, and identifying and recording the most extreme cluster value in the shuffled data. This process was repeated 6,000 times, producing a distribution of cluster values from shuffled data. The cluster values identified in the actual observed data were compared to this permutation distribution, and the reported *p*-values represent the percentile ranking of the observed clusters compared to the permutation distribution. The difference between conditions was considered significant when the observed cluster statistic fell within one of the 2.5th percentiles of the null distribution (reflecting a two-sided test); thus, any *p* value less than 0.025 was considered statistically significant.

Mass univariate tests were conducted to compare conditions of interest for pre-stimulus and post-stimulus oscillatory activity, separately for young adults and older adults. The condition contrasts tested are specified in the Results section. Additionally, to directly test the impact of aging on oscillatory activity (i.e. the interaction of age group and condition), we conducted mass univariate tests on condition differences between age groups. Specifically, differences in oscillatory activity were calculated between conditions (e.g. Expected vs. Unexpected) for each participant in both age groups, and independent *t*-tests were conducted at each time–frequency–channel bin, testing for a significant difference between groups. These tests were corrected with the cluster-based method as previously described.

Correlations of oscillatory responses

We conducted correlation analyses to relate the pre-stimulus oscillatory activity to changes in oscillatory power following the presented word and to investigate if this relationship differed between age groups. For each participant, average oscillatory power was extracted from the cluster of channels, time window, and frequency band determined to be significant by the previously

described cluster-based permutation statistics (i.e. the data were averaged across time, frequency, and channel for the extent of the significant cluster that was identified, resulting in a single value reflecting the average cluster power). These average oscillatory power values were then related with Spearman correlations, as this method is more robust to outliers than Pearson correlations (Pernet et al., 2013; Rousselet & Pernet, 2012). Correlations were calculated between variables for both age groups. Given the number of tests conducted, we corrected the resultant *p*-values with the false discovery rate correction method (Benjamini & Hochberg, 1995). Additionally, to avoid fallacious interpretations of the presence of interactions based on differences in correlation magnitudes (Nieuwenhuis et al., 2011), we directly statistically tested the correlation differences between groups using multiple methods, implemented by the *cocor* package in R (Diedenhofen & Musch, 2015). When reporting statistical differences between correlations, we report Steiger's *z* (Steiger, 1980) and Zou's 95% CI for differences in the correlation (Zou, 2007).

Results

Analysis of pre-stimulus oscillatory power

The analysis of oscillatory activity prior to the critical final word of the sentence collapsed across Expected and Unexpected ending conditions to assess the difference between Strongly Constraining and Weakly Constraining sentence frames. Since participants were unaware prior to the onset of the final word whether it would be an expected or an unexpected completion, there should be no difference between these conditions prior to the onset of the word, and so we focused on how sentential constraint influenced anticipatory oscillatory activity. The results of this analysis are depicted in Figure 1.

In the young adult readers, a significant difference (clusterwise $p = 0.005$) in oscillatory power prior to the onset of the critical word was found between Strongly Constraining and Weakly Constraining sentences (shown in Figure 1(A)). This cluster ranged from roughly -200 – 0 ms prior to the onset of the critical word, and spanned frequencies from 5 to 20 Hz, primarily reflecting a greater decrease in alpha / beta oscillatory activity prior to the onset of a more predictable word. The cluster was centred primarily over posterior channels, with a right posterior maximum. This pattern of results was largely similar to what was initially reported by Rommers et al. (2017).

Interestingly, in the older adult readers, a similar significant difference (clusterwise $p = 0.01$) in oscillatory power as the younger adult readers was found

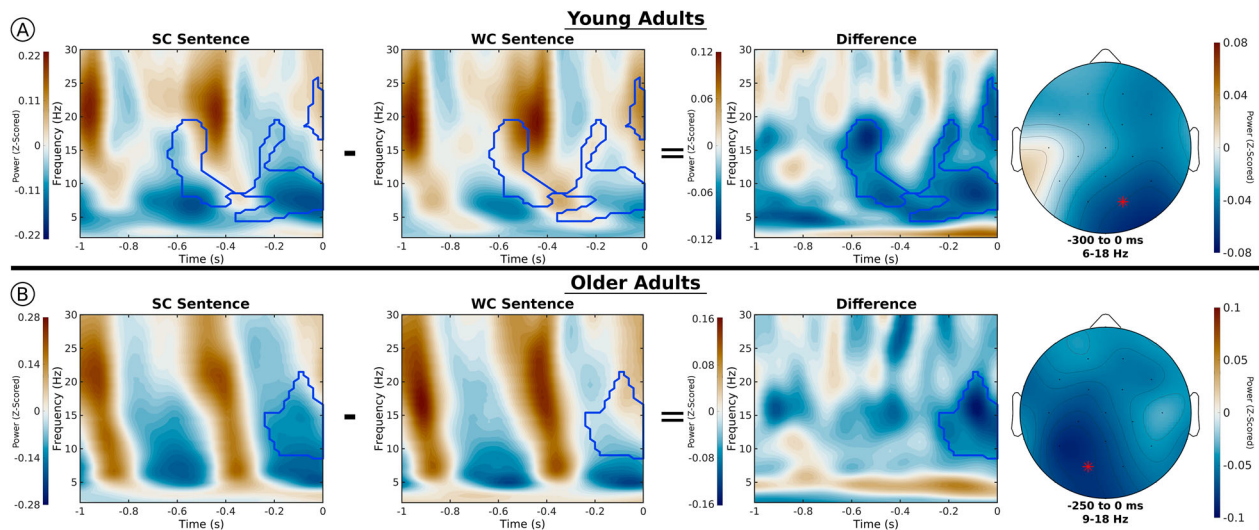


Figure 1. Time–frequency and scalp topography plots depicting significant differences in pre-stimulus oscillatory activity in (A) younger and (B) older adults. The three leftmost plots show, in order, oscillatory power across time (x axis) and frequency (y axis), first for the Strongly Constraining (SC) sentences, then for the Weakly Constraining (WC) sentences, and finally for the difference between the two. The outline on all plots depicts the clusters showing statistically significant power differences. The rightmost plot shows average power across the scalp in the time window and frequency band specified, with the maximal cluster channel (which is the channel plotted in the time–frequency plots) highlighted with a star.

between Strongly Constraining and Weakly Constraining sentences (shown in Figure 1(B)). This cluster ranged from roughly -200 – 0 ms prior to the onset of the critical word, and spanned frequencies from 8 to 18 Hz, primarily reflecting a greater decrease in alpha / beta oscillatory activity prior to the onset of a more predictable word. Although the overall topographies between age groups appeared to differ, the significant cluster identified in the older adults also spanned primarily over posterior channels, with a centre posterior maximum.

We tested the interaction of age group and condition with a mass univariate test on condition differences in oscillatory power (Strongly Constraining – Weakly Constraining) between age groups. This analysis reported no significant interaction (i.e. no significant clusters were found). Thus, both age groups exhibited a decrease in alpha/beta oscillatory activity prior to the onset of a more predictable word, and this decrease did not appear to significantly differ between the age groups.

Analysis of post-stimulus oscillatory power

Analysis of oscillatory activity following the onset of the critical final word of the sentence focused on specific condition differences. Of primary interest was the difference in activity following Expected vs. Unexpected words in Strongly Constraining sentences, in order to examine the engagement of processes following a violation of a strong prediction (although this difference between Expected and Unexpected endings was also

tested in Weakly Constraining sentences). We focused analyses on differences between sentence ending types within constraint conditions in order to control for the preceding sentence context. An analysis of oscillatory power differences between Expected vs. Unexpected words in Strongly Constraining sentences in young adult readers revealed significant differences in power in two clusters. These clusters are depicted in Figure 2.

The first significant cluster (clusterwise $p = 0.0002$; Figure 2) identified significantly greater oscillatory power following Unexpected words compared to Expected words. This cluster was centred primarily over frontal and central channels and spanned from roughly 200 – 700 ms following the onset of the critical word. The cluster primarily spanned frequencies from 3 to 7 Hz, demonstrating an increase in frontal theta activity following Unexpected sentence endings of Strongly Constraining sentences.

The second significant cluster (clusterwise $p = 0.009$; Figure 2) identified a significant reduction in oscillatory power following Unexpected words compared to Expected words. This cluster was centred primarily over left frontal and central channels and spanned from roughly 500 – 700 ms following the onset of the critical word. The cluster primarily spanned frequencies from 9 to 18 Hz, demonstrating a decrease in alpha band activity following Unexpected sentence endings of Strongly Constraining sentences. As before, the pattern of results seen post-stimulus was largely

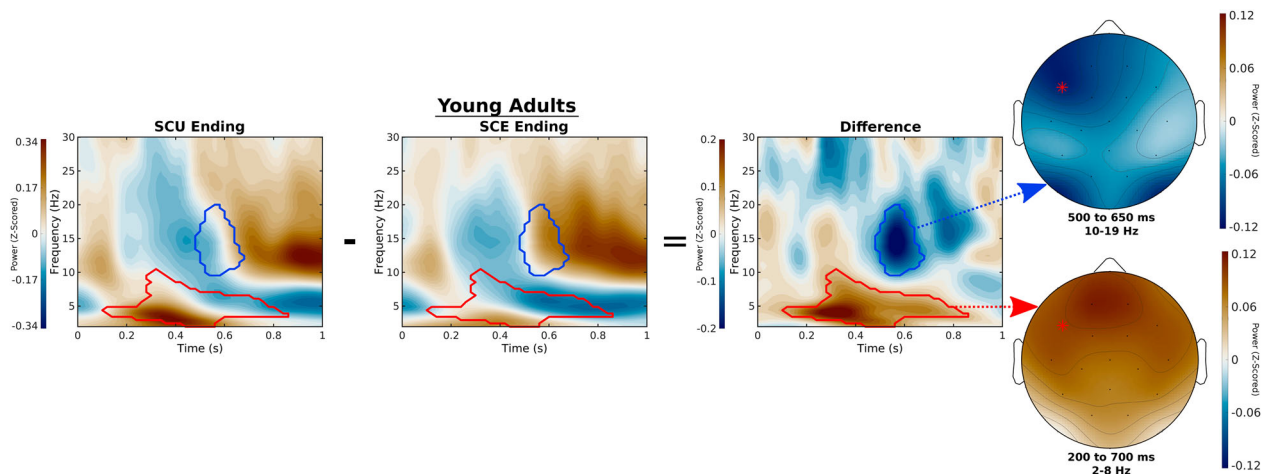


Figure 2. Time–frequency and scalp topography plots depicting significant differences in oscillatory activity following the critical word in younger adults. The three leftmost plots show, in order, oscillatory power across time (x axis) and frequency (y axis), first for the Strongly Constraining Unexpected (SCU) final word condition, then for the Strongly Constraining Expected (SCE) final words, and finally for the difference between the two. The highlighted cluster and topography (at right, bottom) shows the significant positive cluster associated with frontal theta band activity, whereas the highlighted cluster and topography (at right, top) shows the significant negative cluster associated with frontal-central alpha band activity.

similar to what was initially reported by Rommers et al. (2017).

This analysis was also conducted in the group of older adult readers, revealing interesting differences compared to the young adult readers (depicted in Figure 3). Comparing Unexpected and Expected endings to Strongly Constraining sentences, a cluster indicating greater engagement of frontal theta band activity was identified (see Figure 3), but only trended towards significance (clusterwise $p=0.04$), although this cluster reached

significance when a reduced range of frequencies, informed by the younger adult frontal theta result, was entered into the cluster-based permutation test (clusterwise $p=0.002$). Compared to the similar effect in younger adult readers, the frontal theta effect in older adults was delayed, spanning roughly 500–900 ms, and was somewhat lower in frequency, ranging from 2 to 5 Hz. Thus, a frontal theta response was also found in older adult readers, although it was reduced in magnitude and delayed compared to younger adults.

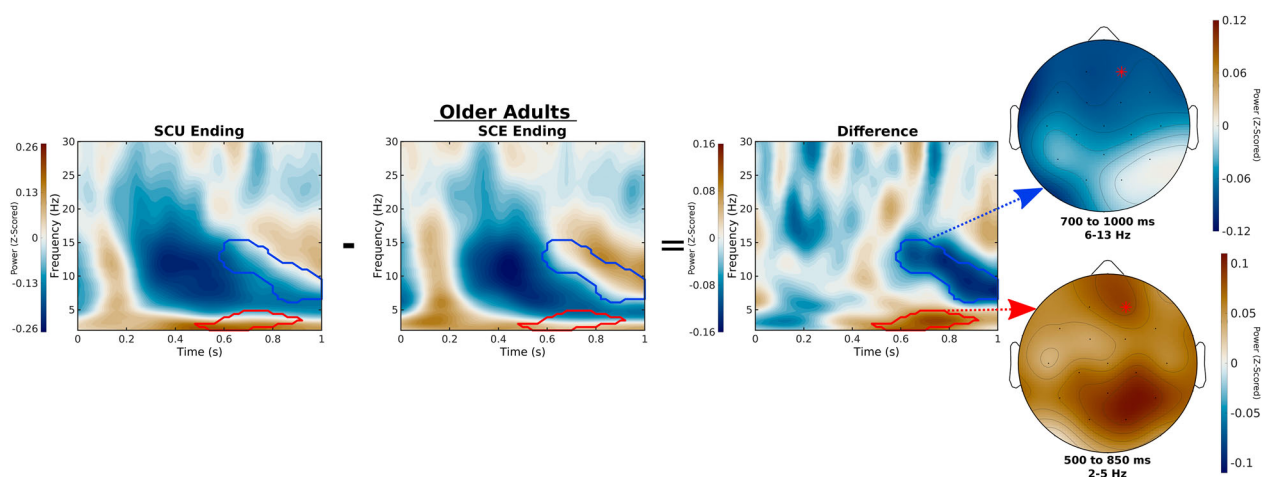


Figure 3. Time–frequency and scalp topography plots depicting significant differences in oscillatory activity following the critical word in older adults. The three leftmost plots show, in order, oscillatory power across time (x axis) and frequency (y axis), first for the Strongly Constraining Unexpected (SCU) final word condition, then for the Strongly Constraining Expected (SCE) final words, and finally for the difference between the two. The highlighted cluster and topography (at right, bottom) shows the significant positive cluster associated with frontal theta band activity, whereas the highlighted cluster and topography (at right, top) shows the significant negative cluster associated with frontal alpha band activity.

The analysis comparing Unexpected and Expected endings of Strongly Constraining sentences in older adult readers also revealed a second significant difference (clusterwise $p = 0.01$; Figure 3). This cluster identified a significant reduction in alpha-band oscillatory power following Unexpected words compared to Expected words. Compared to the similar effect in younger adult readers, the alpha-band activity decrease in older adults was delayed, spanning from 800 to 1000 ms, and was slightly lower in frequency range, ranging from 6 to 12 Hz. Thus, older adults elicited similar responses to younger adults following Unexpected endings, but these responses were delayed and lower in frequency.

As with the analysis of pre-stimulus activity, we tested the interaction of age group and condition in post-stimulus activity between Expected and Unexpected words in Strongly Constraining sentences. The difference in oscillatory power between Unexpected and Expected endings was computed for each group, and the difference of differences between groups was tested statistically. The conducted comparison was young adult condition difference – older adult condition difference, and, as such, a positive cluster in this analysis would reflect a more positive condition difference in the young adult neural data, and a negative cluster would reflect a more negative condition difference. This analysis revealed two significant effects (shown in Figure 4).

The first significant effect (clusterwise $p = 0.02$) was characterised by a positive cluster, indicating a more positive power difference for young adult readers compared to older adults. This cluster spanned from roughly 200–500 ms following the onset of the critical word, and ranged from 2 to 7 Hz, with a topography largely focused over left frontal sites. This largely overlapped with the frontal theta effect observed in young adult readers; thus, young adult readers engaged

frontal theta mechanisms to a greater degree than older adults following Unexpected sentence endings.

The second significant effect (clusterwise $p = 0.02$) was characterised by a negative cluster, indicating a more negative power difference for young adult readers compared to older adults. This cluster spanned from roughly 450–600 ms following the onset of the critical word, and ranged from 10 to 18 Hz, with a topography over left frontal and posterior sites. This largely overlapped with the frontal alpha / beta decrease observed in young adult readers; thus, young adult readers engaged frontal alpha mechanisms to a greater degree than older adults following Unexpected sentence endings. Overall, the theta and alpha responses found in younger adult readers were greater in magnitude compared to older adults. Importantly, no interaction clusters were identified in the time / frequency / channel bins where the delayed older adult theta and alpha responses were found, suggesting the magnitude of these responses did not largely differ from the ongoing activity of younger adults.

Additional tests compared oscillatory activity elicited by Unexpected vs. Expected endings to Weakly Constraining sentences. In young adult readers, a significant difference (clusterwise $p = 0.0003$) was found between these conditions. This difference was similar in timing to the frontal alpha decrease found when comparing Unexpected and Expected endings in Strongly Constraining sentences, although the range of frequencies and topography differed to some extent. Interestingly, no significant differences (lowest clusterwise $p = 0.27$) were found when this comparison was tested in older adult readers; however, a test of the interaction between condition and age group also revealed no significant differences between age groups. In summary, younger adults, but not older adults elicited a decrease in alpha power following Unexpected endings to

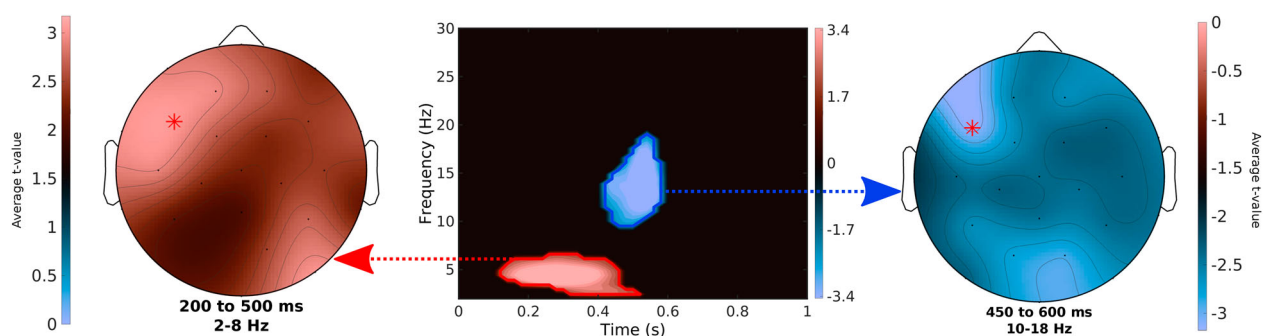


Figure 4. Clusters showing interaction between age group and condition (unexpected – expected endings of strongly constraining sentences). Average t -values from statistical tests are plotted, both in the centre time–frequency plot and the topographic plots of the significant clusters. The positive cluster indicates a more positive power difference for young adults compared to older adults, whereas the negative cluster indicates a more negative power difference.

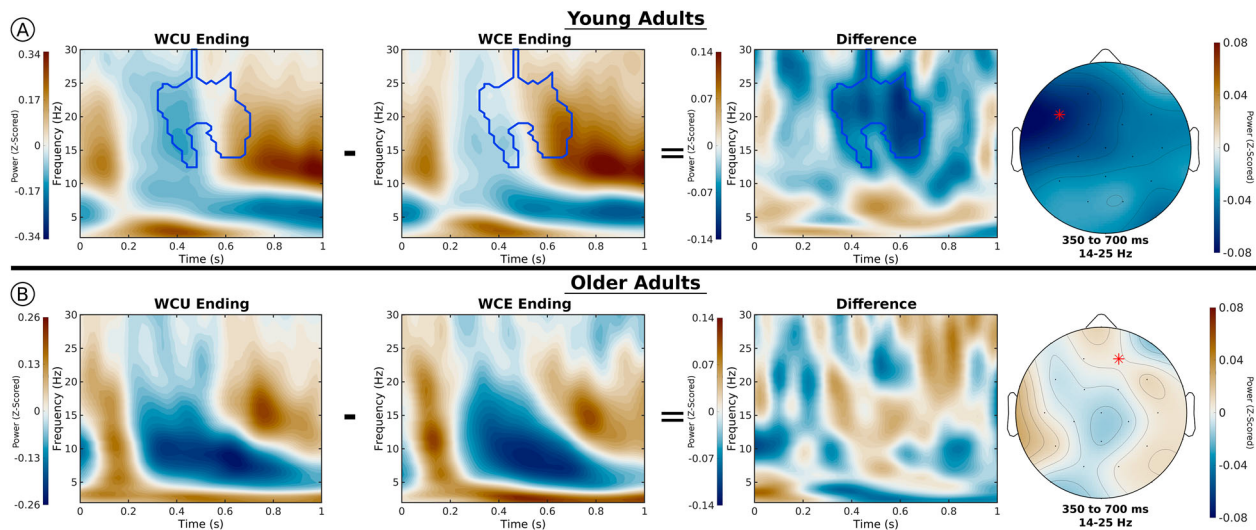


Figure 5. Time–frequency and scalp topography plots depicting significant differences in oscillatory activity following the critical word in weakly constraining sentences in (A) younger and (B) older adults. The three leftmost plots show, in order, oscillatory power across time (x axis) and frequency (y axis), first for the weakly constraining unexpected (WCU) final word condition, then for the weakly constraining expected (WCE) final words, and finally for the difference between the two. The bold outline depicts the statistically significant difference cluster in the young adult data. The right side plot shows average power across the scalp in the time window and frequency band specified, with the maximal cluster channel highlighted with a star. A significant difference was found in younger adults; no significant effect was found in older adults, and the plots are presented for comparison.

Weakly Constraining sentences, but a statistical test of this age group difference was inconclusive. These results are presented in Figure 5.

Given that a frontal alpha decrease was observed following Unexpected endings in both Strongly and Weakly constraining contexts in young adult readers, one possibility is that this effect is related to unexpectedness in general rather than being a more specific prediction violation effect. To statistically test this possibility, a cluster-based permutation test was conducted to compare the difference of condition differences. For each subject, the difference in oscillatory activity between Unexpected and Expected endings of Strongly Constraining sentences, as well as the difference in oscillatory activity between Unexpected and Expected endings of Weakly Constraining sentences, was calculated. The difference of these condition differences (i.e. SCU-SCE vs. WCU-WCE) was statistically tested. In young adult readers, this analysis revealed a significant difference (clusterwise $p = 0.007$), in that a greater frontal theta effect was observed in the Strongly Constraining condition compared to the Weakly Constraining condition. This effect largely overlapped in frequency, time, and topography with the result displayed in Figure 2(A). A similar significant difference was found in the sample of older adult readers (clusterwise $p = 0.022$); namely, a greater frontal theta effect was observed in the Strongly Constraining condition compared to the Weakly Constraining condition. The effect

in the older adult readers was consistent with the result shown in Figure 3(A) in terms of frequency, time, and topography. Importantly, no significant differences were found across sentential constraint for the frontal alpha effect for either young or old adults. Thus, the frontal theta effect – but not the alpha effect – was specific to violations of strong predictions in both young and older adults.

Correlations

To relate pre-stimulus to post-stimulus processing, correlations between measurements of oscillatory activity were computed for both age groups. Since pre-stimulus alpha decreases were found to be greater in Strongly Constraining sentences, and the frontal theta effect was selective to prediction violations in Strongly Constraining sentences as well, we focused the correlation analyses on the effects found in Strongly Constraining sentences. Average power values were extracted from the previously identified statistically significant clusters (i.e. averaged across time, frequency, and channel for the extent of the significant cluster) for each participant, and these average power values were correlated with each other (thus, correlations were computed at the subject level, not at the trial level).

In young adult readers, a significant positive correlation was identified between the magnitude of the pre-stimulus alpha decrease prior to more Expected

sentence endings and the magnitude of the post-stimulus frontal theta increase following Unexpected sentence endings ($r = 0.59$, corrected $p = 0.007$), suggesting a relationship between preparatory activity and processing following a prediction violation. The positive direction of this correlation means that individuals who exhibited greater pre-stimulus alpha decreases also showed less of a frontal theta response following unexpected words. Importantly, this relationship was not significant for older adult readers ($r = -0.04$, corrected $p = 0.84$), and the correlations for young adults and older adults significantly differed (Fisher's $z = 2.49$, $p = 0.013$; Zou's 95% CI = 0.13–1.07). This difference is depicted in Figure 6. All other computed correlations (pre-stimulus alpha with post-stimulus alpha; post-stimulus alpha with post-stimulus theta) were not significant for either age group (all corrected p -values > 0.29). Thus, young adult readers' anticipatory processing was related to their post-stimulus theta activity, but this relationship was not observed in older adult readers.

In the previously described analyses, the mean power values extracted for each age group were from clusters identified in that age group's data (e.g. for young adults, the frontal theta cluster was the cluster shown in Figure 2(A)). One possibility is that, although the significant clusters of condition differences identified in the older adults' data differed from those in the young adults' data, there may be variability in the older adults' data in the time, frequency, and channel clusters that were found in the *young adults'* data that is important, or vice versa. We therefore also computed correlations between measurements of mean oscillatory power extracted from previously identified statistically significant clusters, but from the opposite age group

(e.g. statistically masked the older adult data with the young adult clusters). No correlation computed following this method reached statistical significance (all corrected p values > 0.29), suggesting that the relationship between pre-stimulus activity and post-stimulus frontal theta found in the young adults was selective to this age group.

Discussion

Our brains undergo many structural and functional changes as we age, and these changes can have important consequences on our cognitive abilities, as well as the engagement of particular neurophysiological mechanisms during certain tasks. Prior work has demonstrated that aging can impact specific language comprehension abilities, such as the construction and use of message-level information from a sentence context (Federmeier et al., 2007; Payne & Federmeier, 2018), as well as the engagement of anticipatory processing to predict features of upcoming words (DeLong et al., 2012; Federmeier et al., 2002). To study differences in processing due to age, many studies have employed EEG recording methods to understand the temporal dynamics of neural processing related to language comprehension. However, these studies have focused primarily on differences in the amplitude of certain ERP components, such as the N400. In the current study, we sought to expand on this growing literature by instead examining changes in oscillatory activity during language processing and how these mechanisms are impacted by aging. By re-analysing two studies with the same materials, we were able to make controlled comparisons between age groups in order to shed new light on the impact of aging on oscillatory neural dynamics related to predictive processing.

We first analysed differences in oscillatory activity prior to the onset of a word that, given the constraint of the sentence context, was likely to be highly predictable vs. less predictable, in order to examine anticipatory processing related to an upcoming word. Both younger and older adult readers showed a reliable decrease in alpha / beta band oscillatory power prior to the onset of a more constrained sentence ending. While aspects of the topography and specific frequencies of this effect did differ somewhat between age groups, we note that comparing the exact spatial or spectral extent of significant clusters may be inappropriate (Sassenhagen & Draschkow, 2019) and, more generally, that topographic differences are not unexpected and are always difficult to interpret in between-subjects comparisons. Given the more general similarity in the patterns and the lack of a reliable age effect in the

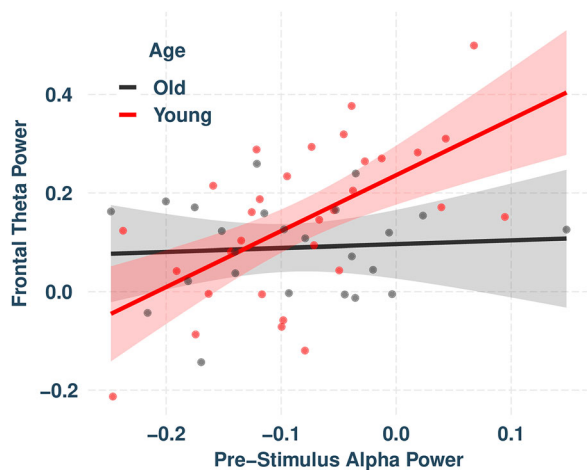


Figure 6. Correlations between pre-stimulus alpha power and post-stimulus frontal theta power for both younger and older adult readers.

statistics, we interpret this to be the engagement of the same mechanism across age groups. This result was surprising, as previous work has suggested that, compared to younger adults, older adults are less likely to show effects of predictive processing prior to a predictable word (DeLong et al., 2012). More specifically, when young adults were presented with sentences such as “the boy went outside to fly *a kite*”, if the critical ending was instead replaced with “*an airplane*”, a larger N400 response was observed following the article “*an*”, suggesting that the prediction for the word “*kite*” was already in place at the time of the article, demonstrating pre-activation of specific lexical items (DeLong et al., 2005). In contrast, older adult readers did not show any N400 amplitude differences at the time of the article, only at the noun, suggesting a lack of specific lexical pre-activation (DeLong et al., 2012). One possible explanation of the discrepancy between these N400 results and the current pre-stimulus alpha results is that the anticipatory mechanism indexed by the alpha / beta decrease may differ from other anticipatory processes, and that this particular mechanism is preserved over the lifespan. For instance, studies using simple spatial cuing tasks have reported similar alpha power decreases prior to a predictable stimulus in older adult participants compared to younger adults (Heideman et al., 2018). This alpha power decrease may be more of a domain-general anticipatory response, and not as related to the pre-activation or preparation of specific linguistic features of upcoming stimuli, although recent work suggests a possible link between fluctuations in alpha-band activity and language-related ERP amplitudes (Lago et al., 2023). As such, this anticipatory neural signal may be relatively preserved in older adult readers, whereas specific lexical prediction may be diminished.

In contrast to the pre-stimulus effects, differences between age groups emerged when examining the oscillatory mechanisms engaged following prediction violations compared to predictable sentence endings. Both younger and older adult readers elicited an increase in frontal theta power and a decrease in alpha band activity following prediction violations of strongly constraining sentences; however, older adult responses were temporally delayed, lower in frequency, and smaller in magnitude. Interaction analyses suggested that the responses engaged by young adults were greater in magnitude than older adults, whereas the older adult responses were not significantly greater than the ongoing activity of the young adults’ neural data. Importantly, focused statistical comparisons revealed that, for both young and older adults, the frontal theta effect was specific to strongly constraining

contexts, suggesting not only that this response is linked to the engagement of control following prediction violations, but also that the delayed theta response observed in older adult readers may still reflect the same control process as young adult readers. The frontal alpha decrease may reflect a more general response following unexpected events, as this response was present in both Strongly and Weakly Constraining contexts in young adult readers. However, older adults showed no significant differences in oscillatory power when comparing predictable and unexpected endings of weakly constraining sentences, a pattern similar to what has been observed with the N400 (Wlotko et al., 2012). It remains an open question whether the frontal alpha decrease is a more graded response, and the effect was simply not strong enough to observe in weakly constraining contexts in older adults, or is a more state-like response that older adults do not engage when the context is not constraining.

The current results suggest that older adults may not engage post-error control processes to the same degree as younger adults, or at the very least that engagement of these processes is delayed and that these processes may only be engaged when the context is highly constraining. Interestingly, parallel work examining ERPs to unexpected words suggests older adults are also less likely to elicit post-N400 frontal positivity ERP responses following linguistic prediction violations of strongly constraining sentences (Federmeier et al., 2010; Wlotko et al., 2012). These responses may reflect the engagement of revision processes to update the comprehender’s situation model with the unexpected information (Kuperberg et al., 2020). Thus, one possibility is that older adults may not engage control processes following prediction violations to the same degree as younger adults, leading to less revision or prediction updating downstream. When constraint is low, older adults may not engage frontal alpha decreases, and when constraint is stronger, their frontal positivity ERP responses are reduced, and the frontal theta responses are delayed and reduced in magnitude.

Correlation analyses relating the pre-stimulus alpha decrease to post-violation theta activity provided additional support for the notion that older adults do not engage in control or revision processes following prediction violations. Namely, the engagement of anticipatory alpha decreases was related to post-violation theta power in younger adult readers, but this relationship was not present in older adults. Rommers et al. (2017) speculated that this relationship might reflect a strategic trade-off, such that a greater degree of anticipation of the upcoming stimulus (i.e. a greater alpha decrease) may have mitigated the need for post-violation

control processes (i.e. a reduced frontal theta response), although this correlation was at the subject level, not the individual trial level. In older adults, the degree of anticipation of an upcoming stimulus appeared to have no link to their future frontal theta response. One possibility is that older adults may engage in some anticipatory processing but are less able to link their predictions to their mental situation model or to update / revise their model following prediction violations. For instance, when upcoming words are presented in the parafovea, younger adults elicit N400 amplitude differences to semantic violations to parafoveally presented words, but this congruency effect is mitigated when the word reaches the fovea, suggesting that young adults can extract semantic information from the upcoming word, and integrate this information by the time the word is apprehended in foveal vision (Payne et al., 2019; Stites et al., 2017). Older adults also show N400 congruency effects to words presented parafoveally; however, when these words are apprehended foveally, the congruency effect is not mitigated, suggesting the information from the parafovea was not integrated into the ongoing representation of the sentence (Payne & Federmeier, 2017). Here, similarly, older adults may have engaged anticipatory processes to some degree, reflected by the pre-stimulus alpha decrease, but then not have been able to make use of this preparatory activity in their later processing, once the word was presented.

Older adults' oscillatory responses following prediction violations were delayed and lower in frequency compared to younger adults. This may be due to cognitive differences between age groups, but physiological changes across age could also explain these differences, as numerous results suggest overall slowing of brain activity over the lifespan (Courtney & Hinault, 2021). Prior research has also demonstrated that oscillatory rhythms shift with age; for instance, higher frequency activity decreases over time, whereas lower frequency activity increases (Duffy et al., 1993). Alpha power and peak alpha frequency tend to decrease with age (Klimesch, 1999), particularly over parietal and occipital sites, and this effect is observed above and beyond the flattening 1/f aperiodic component that is observed with aging (Merkin et al., 2023; Tröndle et al., 2023). Additionally, onset and peak latencies of numerous ERP components in different tasks exhibit delays with aging (Curran et al., 2001; Kropotov et al., 2016; Kutas & Iragui, 1998), and onset delays have also been observed for changes in oscillatory power (Deiber et al., 2013; Zanto et al., 2010). Although these delays are not usually of the magnitude observed in the current study, it is possible that the differences in post-violation responses between younger and older adults

reflect more fundamental neurophysiological changes. Nevertheless, the lack of a relationship between pre-stimulus and post-stimulus responses specifically in older adult readers suggests these differences between age groups have important implications for cognitive processing as well.

The current results suggest that older adults exhibit altered oscillatory responses following prediction violations compared to young adult readers; however, this work has some limitations, in that other potential explanatory factors of age-related differences were not measured. Other work has identified that "age" as a construct may be less important at predicting neural changes than the individual differences that develop over the course of the lifespan (Bornkessel-Schlesewsky et al., 2015). For instance, in studies examining language-related ERPs, older adults with high verbal fluency scores show responses that are more similar to young adult participants (Federmeier et al., 2002, 2010). In another study, older adults with higher performance on cognitive control tasks showed greater N400 effects of predictability in a language task (Dave et al., 2021). Other work suggests that older adults with greater working memory abilities may show greater engagement of prediction when processing language (Huettig & Janse, 2016; although see Jongman et al., 2022). Individual differences in reading habits or print exposure over the lifespan may also have important consequences for language comprehension and processing (Payne et al., 2012). Further work will be necessary to identify potential cognitive factors that might explain age-related changes in oscillatory responses related to prediction during language comprehension.

In conclusion, the current study examined changes in oscillatory activity during language comprehension in both young and older adults. Our results suggest that older adult readers may engage some similar or preserved anticipatory processes compared to young adult readers, as both age groups elicited alpha power decreases of a similar magnitude prior to more predictable words. In contrast, older adults' elicitation of frontal theta control processes following prediction violations was delayed and reduced in magnitude compared to young adults, and, different from patterns in the young, had no relationship with the aforementioned pre-stimulus alpha decreases. These results contribute to growing literature characterising age-related shifts in language processing, and potentially suggest that older adults may not show impairments in all forms of prediction, but may show specific impairments in linking the contents of their predictions to the developing mental situation model, and/or in revising their predictions following a prediction violation.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data that support the findings of this study are available on request from the corresponding author. The conditions of our ethics approval at the time the data were collected do not permit public archiving of the EEG data in this study due to risk of identification of individuals through biological signals. Access will be granted to named individuals in accordance with ethical procedures governing the reuse of sensitive data, including completion of a formal data-sharing agreement and approval by the local ethics committee. However, the analysis code used for this study is provided at <https://osf.io/4wyc9/>.

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